Simoa blood biomarkers (ptau181 analysis at Gothenburg) tested on Insight46 phase 1 plasma and serum samples from the LWBL/UKDRI biobank

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Document details

Categories of variables	Simoa blood biomarkers tested on phase 1 plasma and serum samples from the LWBL/UKDRI biobank
Summary of work undertaken	Source data transferred to Stata file Missing data – labelled as -99. Methods section below
Names of source variables	SubjectID, Age_at_blood, Sex, plasma_available_ptau181_Got, ptau181_plasma_p1_Got
Output variables	age_blood_i46, ptau18_plsma_av_g_i46, ptau18_plsma_g_i46
Papers using these variables	1. Keshavan A, Pannee J, Karikari TK et al. Population-based blood screening for preclinical Alzheimer's disease in a British birth cohort at age 70. Brain. 2021 Jan 22. pii: 6105867. doi: 10.1093/brain/awaa403.

Methods

General assay methods are described in Karikari et al 2020, Lancet Neurology.

For plasma p-tau181 quantification in Insight 46, specific methods are taken from reference 1 above as follows: a 0.5 mL aliquot of plasma for each individual was transported on solid carbon dioxide to the University of Gothenburg, Sweden; samples were immediately stored at -80 C until use. On the day of the analysis, the blood samples were allowed to thaw at room temperature for an hour, vortexed at 200 rpm for 30 s, and centrifuged at 4000 g for 10 min, then 90 µL of the supernatant was used for p-tau181 measurements. Separate anonymized plasma samples from the University of Gothenburg, pooled to give two different p-tau181 concentrations, were used as internal quality control (iQC) samples, and were processed identically to the test samples. Each iQC sample was analyzed in duplicate at the start and the end of each plate. Combined iQC data from eight separate runs were used to determine the within- and between-run variations, following the recommendations of an international group of neurochemists (Andreasson et al., 2015).

The high-concentration iQC sample (concentration expressed in mean, standard deviation =17.1, 0.7 pg/mL) had a within-run variation of 0.4% and a between-run variation of 10.3%.

For the low-concentration iQC sample (concentration = 4.8, 0.5 pg/mL), the within- and between-run variations were 10.9% and 13.0% respectively. Identical batches of assay reagents were used throughout the measurements for this cohort.

Notes on sample availability for analysis Where no aliquot of plasma or serum was available in the biobank for a given individual (due to aliquots having been used up or failure of blood sampling or incomplete blood sampling) the variable plasma_available_ptau181_Got indicates this. For the blood biomarker variables, the code -99 indicates no quantification was performed.

It is recommended that prior to making a request for these variable researchers discuss the plan with Dr Ashvini Keshavan.

Variables in this document

3 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Biological samples [6037] › Blood biomarkers [60372]		
age_blood_i46	Age at blood sampling (years) - Insight46 phase 1	topsheet field
ptau18_plsma_av_g_i46	Plasma available for Simoa ptau181 analysis at Gothenburg - Insight 46 phase 1	topsheet field

Variable	Label	How it was found
ptau18_plsma_g_i46	Plasma p-tau181 Gothenburg homebrew, pg/ml - Insight 46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan

1 low-confidence match

Variable	Label	How it was found
Population characteristics [54] › Baseline characteristics [401]		
sex	Sex of the participant	prose scan